PRODUCT USER MANUAL

For

Mediterranean Ocean Heat Content Anomaly (0-700m)

OMI_CLIMATE_OHC_MEDSEA_area_averaged_anomalies

Issue 1.4

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1.4	28-May-2025	all	Updated template, Updated temporal extension, setup of operational extension and renaming of the product name	Rafael Gomes de Menezes	Emanuela Clementi	Copernicus Marine Product Management

GLOSSARY AND ABBREVIATIONS

ARMOR	ARMOR3D insitu observational dataset
CGLOR	C-GLORS reanalysis
CORA	CORA5.2 insitu observational dataset
CMEMS	Copernicus Marine Environment Monitoring Service
FOAM	FOAM Reanalysis
GLO	Global
GLOR	GLORYS2V4 reanalysis
GREP	Global Reanalysis Ensemble Product
MED	Mediterranean
MEDREA	Mediterranean Sea (Med-MFC) Reanalysis
MFC	Monitoring and Forecasting Center
MOI	Mercator Océan International
NetCDF	Network Common Data Form
OHC	Ocean Heat Content
OMI	Ocean Monitoring Indicator
ORAS	ORAS5 reanalysis
PUM	Product User Manual
QUID	Quality Information Document
TAC	Thematic Assembly Center

Table of Content

GLOSSARY AND ABBREVIATIONS 3

DATA ACCESS 5

I INTRODUCTION 5

I.1. Summary 5

I.2. History of changes 6

II DESCRIPTION OF THE PRODUCT 6

II.1. General information 6

II.2. Details of the dataset 7

II.3. Product System Description 7

III FILES NOMENCLATURE AND FORMAT 8

III.1. Nomenclature of files 8

III.2. File Format 8

IV REFERENCES 8

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

I INTRODUCTION

I.1. Summary

This document is the user manual for the Copernicus Marine Service Ocean Monitoring Indicator (OMI) OMI_CLIMATE_OHC_MEDSEA_area_averaged_anomalies. It provides a time series of Ocean Heat Content (OHC) anomalies area-averaged over the Mediterranean Sea (30°N, 46°N; 6°W, 36°E), between 0 and 700 meters depth, excluding regions where the seafloor is shallower than 300 meters (see the QUID document for more details on the calculation method).

The product consists of a one-dimensional (time) dataset with annual mean of OHC anomalies, computed from a combination of reanalysis and observational products: MEDREA[1], CGLOR, ORAS, FOAM reanalysis (from the GREP [2] ensemble), and CORA5.2 [3] and ARMOR3D [4] observational temperature data.

The time series begins in 1993 and, for each product, all available data is used, resulting in a varying time coverage. Consequently, the ensemble is calculated over the time period common to all products, extending at least up to three years prior to the current year [−3Y] (or closer to the present, depending on data availability), and it is extended operationally on an annual basis. For the MEDREA, both reanalysis and interim data are used, thus the index is calculated up to the year preceding the current one [-1Y].

The product is published on the Copernicus Marine Service dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the Copernicus Marine Service Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

[1] MEDSEA_MULTIYEAR_PHY_006_004: DOI: https://doi.org/10.25423/CMCC/MEDSEA_MULTIYEAR_PHY_006_004_E3R1
[2] GLOBAL_REANALYSIS_PHY_001_031
[3] INSITU_GLO_TS_REP_OBSERVATIONS_013_001_b
[4] MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012

I.2. History of changes

2018/09	Publication of MEDSEA_OMI_OHC product: OMI_CLIMATE_OHC_MEDSEA_area_averaged_anomalies
2021/01	Time extension for year 2018 of OMI_CLIMATE_OHC_MEDSEA_area_averaged_anomalies using the new Med Sea Reanalysis MEDSEA_MULTIYEAR_PHY_006_004 and GLOBAL ENSEMBLE Reanalysis at ¼ degree GLOBAL_REANALYSIS_PHY_001_031
2022/06	Time extension for year 2019
2025/05	Temporal extension and Setup of operational extension

II DESCRIPTION OF THE PRODUCT

II.1. General information

Product name	OMI_CLIMATE_OHC_MEDSEA_area_averaged_anomalies
Geographical coverage	30°N, 46°N; 6°W, 36°E
Variables	Ocean Heat Content (OHC) anomaly
Update frequency	Annual
Available time series	Ensemble -> from 1993 up to at least three years before the current year [-3Y] (or closer to the present, depending on data availability) MEDREA -> from 1993 up to one year before the current year [-1Y]
Delivery mechanism	MDS
Temporal and Spatial resolutions	Annual averages 1/24 degrees
Integration depth	0-700 m, excluded regions shallower than 300 m
Format	NetCDF CF1.7

II.2. Details of the dataset

omi_climate_ohc_medsea_area_averaged_anomalies contains the following variables:
ohc_mean **[J m-2]** **long_name: "Ensemble Mean of Area Averaged Ocean Heat Content Anomaly"** **standard_name: "integral_wrt_depth_of_sea_water_temperature_expressed_as_heat_content"**
ohc_std **[J m-2]** **long_name: "Ensemble Standard Deviation of Area Averaged Ocean Heat Content Anomaly"** **standard_name: "integral_wrt_depth_of_sea_water_temperature_expressed_as_heat_content"**
ohc_glor **[J m-2]** **long_name: "Area Averaged Ocean Heat Content Anomaly from GLORYS2V4 Sea Water Potential Temperature"** **standard_name: "integral_wrt_depth_of_sea_water_temperature_expressed_as_heat_content"**
ohc_cglo **[J m-2]** **long_name: "Area Averaged Ocean Heat Content Anomaly from C-GLORS05 Sea Water Potential Temperature"** **standard_name: "integral_wrt_depth_of_sea_water_temperature_expressed_as_heat_content"**
ohc_oras **[J m-2]** **long_name: "Area Averaged Ocean Heat Content Anomaly from ORAS5 Sea Water Potential Temperature"** **standard_name: "integral_wrt_depth_of_sea_water_temperature_expressed_as_heat_content"**
ohc_foam **[J m-2]** **long_name: "Area Averaged Ocean Heat Content Anomaly from FOAM/GloSea Sea Water Potential Temperature"** **standard_name: "integral_wrt_depth_of_sea_water_temperature_expressed_as_heat_content"**
ohc_cora **[J m-2]** **long_name: "Area Averaged Ocean Heat Content Anomaly from CORA5.2 Sea Water Temperature"** **standard_name: "integral_wrt_depth_of_sea_water_temperature_expressed_as_heat_content"**
ohc_armor **[J m-2]** **long_name: "Area Averaged Ocean Heat Content Anomaly from ARMOR-3D Sea Water Temperature"** **standard_name: "integral_wrt_depth_of_sea_water_temperature_expressed_as_heat_content"**
ohc_med **[J m-2]** **long_name: "Area Averaged Ocean Heat Content Anomaly from Med Sea Water Potential Temperature"** **standard_name: "integral_wrt_depth_of_sea_water_temperature_expressed_as_heat_content"**

II.3. Product System Description

Data extracted from MEDREA and GREP reanalysis[5] or observations[6] provide the temperature which is used to compute the series of OHC anomalies. The annual mean time series start on 1993, each yearly mean is centred on the 30th of June. This product includes annual mean files of OHC temporal series for the Mediterranean Sea (30°N, 46°N; 6°W, 36°E). These averages are calculated from upper 700m depth.

[5] MEDREA: MEDSEA_MULTIYEAR_PHY_006_004; GREP: GLOBAL_REANALYSIS_PHY_001_031
[6] CORA5.2: INSITU_GLO_TS_REP_OBSERVATIONS_013_001_b and ARMOR3D: GLOBAL_REP_PHY_001_021

III FILES NOMENCLATURE AND FORMAT

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center

III.1. Nomenclature of files

OMI_CLIMATE_OHC_MEDSEA_area_averaged_anomalies file nomenclature when downloaded through the Copernicus Marine Service Download Services is based on the following nomenclature:

<datasetname>.nc

For example: omi_climate_ohc_medsea_area_averaged_anomalies.nc

III.2. File Format

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link: What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article: how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IVREFERENCES

Quality Information Document (QUID): https://documentation.marine.copernicus.eu/QUID/CMEMS-OMI-QUID-MED-OHC-area-averaged-anomalies.pdf

Escudier, R., Clementi, E., Omar, M., Cipollone, A., Pistoia, J., Aydogdu, A., Drudi, M., Grandi, A., Lyubartsev, V., Lecci, R., Cretí, S., Masina, S., Coppini, G., & Pinardi, N. (2020). Mediterranean Sea Physical Reanalysis (CMEMS MED-Currents) (Version 1) [Data set]. Copernicus Monitoring Environment Marine Service (CMEMS). https://doi.org/10.25423/CMCC/MEDSEA_MULTIYEAR_PHY_006_004_E3R1